

Can analogue minds live forever?

Introducing the 'Step-Structure Principle' as a testable limit on the Church Turing thesis for brain emulation

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Abstract:

Progress in neuroscience and AI has renewed interest in whether human-level consciousness could be instantiated in non-biological systems, including via whole-brain emulation. A common assumption underlying such proposals is that any function sufficient for consciousness can be implemented on a Turing-equivalent digital computer.

This paper challenges the strength of that assumption by introducing the Step-Structure Principle: while Turing-equivalent architectures can emulate the input–output behaviour and causal logic of any computable function, they do not in general preserve the *step-structure* of a computation: the number and organisation of inputs and outputs involved in the architecture’s computational steps, as well as subsequent aggregate metrics, such as total steps and information processing density.

I argue that this distinction matters if specific architectural features of computation are relevant to phenomenal consciousness. Three independent motivations are developed: (i) constraints arising from computational cost and irreducibility in complex systems, (ii) phase-transition-like thresholds suggested by empirical neuroscience, and (iii) the problem of phenomenal binding in conscious experience. Each motivation is uncertain, but any one would be sufficient to undermine the assumption that digital architectures automatically suffice for human-style consciousness.

The paper outlines empirically testable research pathways to assess whether step-structure is relevant, drawing on advances in neuroscience, computational modelling, and adversarial collaborations in consciousness research. The results bear directly on AI consciousness, digital sentience, and the safety of proposed trajectories toward artificial or uploaded minds.

Introduction

Recent advances in artificial intelligence, neuroscience, and large-scale computational modelling have intensified debate over whether human-level consciousness could, in principle, be instantiated in artificial systems. These discussions arise not only in philosophy, but increasingly in AI research itself, where questions of digital sentience, moral status, and architectural sufficiency have moved from speculative thought experiments to near-term practical concerns.

The current approach to mind uploading has two key assumptions. The first can be called 'physical functionalism' about consciousness, the claim that consciousness is a function of what the brain does as a physical object. The second is 'Turing-equivalence', the claim that any function can be adequately implemented in a Turing Machine equivalent architecture, including a large-enough digital computer. These positions are relatively popular, even if no consensus dominates.ⁱ As our lives are increasingly digitised and AI increasingly impressive, the draw towards these ideas will only grow stronger, as does the need to subject them to scrutiny.

The first claim goes head-to-head with many popular spiritual views of the self – and it has generated centuries of debate, to be touched on in Section 1. However, the claim garners support, in part because the opposite can seem implausible. If complex human-style consciousness has no function, it is hard to understand why natural selection appears to have found value in it.

This paper grants physical functionalism for the sake of argument. I take on the second claim as a more challenging target. Turing-equivalence, embodied in the Church-Turing thesis, is a foundation stone in the computer science tradition and transhumanist literature. The few arguments against Turing-equivalence in the context of human consciousness have drawn on exotic mathematical ideas, but most set a high burden of proof for these arguments and remain unconvinced. Instead, I draw attention to a chink in the Church-Turing thesis as it has been applied in the consciousness debate.

After Section 1 introduces mind uploading, a definition of consciousness, and the debate around physical functionalism, Section 2 asks what counts as 'adequate' implementation for the function of human minds. I argue that, even if the causal logic of *any* function can be mimicked in *any* Turing-equivalent architecture, different architectures do not typically retain the exact nature of the computational steps in their underlying algorithmic languages. Call this aspect of a calculation its 'step-structure' and call the corresponding claim about Turing-equivalence the 'Step-Structure Principle'.

The issue now hangs on whether step-structure might plausibly matter for human-style consciousness. Effectively, what if our minds are analogue rather than digital, in that 'how' they calculate matters more than 'what' they calculate? The 'what' can be emulated on any Turing Machine architecture, including digital computers; but the 'how' of 'step-structure' is not necessarily preserved.

While an open question, Section 3 provides three reasons that this possibility deserves serious consideration, grounded respectively in the computational cost of complex systems, phase transitions in empirical neuroscience, and the phenomenal binding problem. Each reason is uncertain, but any one of them succeeding would suffice to validate concerns about the current technological pathway to mind uploading. Fortunately, the ideas are amenable to empirical testing in the same spirit as recent adversarial collaborations in consciousness research. Section 4 outlines the kind of research pathway that could bring clarity to this crucial question. This research is expensive and ambitious, but worthwhile given the nature of what is at stake. If we get it right, immortality beckons. If we get it wrong, we risk mass suicide.

ⁱ For instance, among philosophers, functionalism was the most popular account of consciousness in a 2020 survey, but still a minority position.^[2] Meanwhile, 25% of AI researchers thought that some AI systems would be conscious by 2034, as did 30% of the public.^[3] The implication for mind uploading is direct: if an AI system can be conscious in its own right, embedding a given human's consciousness in a machine is plausibly just a matter of getting the technology right.

Why does this matter?

If any argument for step-structure relevance holds, then your digitally uploaded mind might be behaviourally identical to your biological mind today. Observers might agree that it talks and acts the same as you. But there would be no sophisticated consciousness inside it. Of course, no-one would know this for sure except for you – and you wouldn't be around to tell anyone. As a modern society, we have proved willing to swap privacy for cheap digital services and to swap biodiversity for cheap energy. I fear that we might also prove willing to swap our minds for the mirage of an eternal sunset – or rather, be seduced into doing so without first checking it really works.

The argument also matters because of the closely related issue of digital sentience, arguably humanity's latest moral test – and perhaps our last. There is concern that a novel moral catastrophe might be unfolding today. Given our limited scientific grasp of consciousness and the rapidly improving capabilities of modern AI systems, we cannot rule out that digital machines will become conscious. Indeed, leading AI labs and academics have placed the possibility that today's large language models or their near-term successors are already conscious at 10% or higher (e.g. Anthropic, Dave Chalmers).^[4] Our argument identifies a new direction for empirical research to assess strict indicators for *human-style* complex consciousness inside any purely digital architecture. Of course, we would still be unable to rule out utterly alien conscious experiences: a vast 'terra incognita' of minds might be possible, as Murray Shanahan from Google DeepMind has argued.^[6] In practical terms, however, until we have a grip on the types of conscious experience we encounter daily and where we largely trust each other's self-reports, then commentary on other conscious phenomena is restricted to emotive speculation.

A philosophical aside

Many have questioned the desirability of radically extended lifespans or digital immortality. Literary and philosophical traditions have often framed immortality as a burden rather than a gift. Eternal youth becomes mired with horror in Oscar Wilde's *Picture of Dorian Gray*. The burden of immortality is prominent in Jorge Luis Borges' *The Immortal*. Even vampires are often depicted as struggling with immortality.

But it would take a confident philosopher to reject a longer life – or at least the chance to choose one. Transhumanists merely extend the already popular choice of using technology to enhance the human condition.^[1] Taken to its natural endpoint, technology will allow us incredible levels of control over our longevity, our capabilities, and even our moods, at least if mind uploading delivers on its promise to recreate our memories, personality, and consciousness inside a machine.

In practice, the longevity on offer through mind uploading technology is contingent and controllable, deflating many of the arguments against the tedium of eternity without the possibility of escape. This is no risk-free existence. The relevant infrastructure might get damaged or your peers might not be relied upon to reboot you. Uploaded minds could still reasonably pursue new technologies to enhance themselves or their survival chances even further. They could still build alliances with other minds, with all the motivating intrigue that comes with it.

There remain arguments against an arbitrarily extended life. Such philosophers as Bernard Williams^[7], Susan Wolf^[8], and Martha Nussbaum^[5] have suggested that our finite limited natures shape our values and the meaning of life. If we could live forever, whatever we become would be a long way from human. Without the undeniable hand of death to give the younger generation their turn, infinite life

also risks an ossification of culture and leadership. There may also be limits on having children; if everyone lives forever, even future technologies might eventually fall foul of scarcity. Eternal physical life also means foregoing the miracles of any afterlife we might believe in.

This paper puts these arguments to one side and allows the transhumanists their dream. If technology can preserve our consciousness in a machine, then perhaps humanity's most hubristic of ambitions, most Faustian of bargains might be within reach: immortality. My worries in this paper concern the current technological pathway towards such eternal life.

So far, I've asked you to suspend judgement about the technologies that might enable mind uploading. Section 1 begins by providing evidence that these possibilities are not mere fantasy.

Section 1. The technical and philosophical context for mind uploading

1.1. Technical viability

Mind uploading – also known as whole brain emulation – is the hypothetical process of scanning a human brain in detail and recreating both its current state and causal interaction structure inside a computer. The computer would run a simulation of the brain's information processing, such that the uploaded mind responds the same way as the original biological brain in every functional sense. The goal is to transfer the personality, memories, and consciousness of an individual from their organic brain to a digital medium.

The idea has been taken seriously. Anders Sandberg and Nick Bostrom published an influential 130-page technical roadmap in 2008.^[15] Scientific journals have run special issues on the topic, such as the 2012 edition of the *International Journal of Machine Consciousness*, edited by Ben Goertzel, an influential figure in AI and robotics.^[11] Parts of the Silicon Valley elite endorse the concept. Dedicated organisations have been established to pursue the technology (e.g. the 2045 Initiative; the Carboncopies Foundation) and to leverage its future success through present-day brain preservation (e.g. Nectome; the Brain Preservation Foundation).

With humanity's remarkable, if traumatic, history with technology and its recent acceleration, it seems unwise to bet against the technical feasibility of brain mapping/simulation – although we note a suspiciously common correlation between a transhumanist's life expectancy and their proposed timelines by which this feat is to be achieved.ⁱⁱ

Neuron-level brain maps have come a long way since the 300-400 neurons in a nematode worm mapped in the 1980s.^[17] In the 2020s, we have mapped up to about 140,000 neurons in a fruit fly (the FlyWire project^[14]) and one cubic millimetre of mouse visual cortex (the MICrONS consortium^[16]). One cubic millimetre may not sound much, but it covered some 200,000 cells and 500 million connections of detailed functional mapping.

There remains a long path towards one hundred trillion connections in the human brain, but Jun Igarashi analysed computer capability projections to suggest full connectome simulation in some

ⁱⁱ Not all estimates fit this pattern, but – being somewhat tongue in cheek – those from Dmitry Itskov, Anders Sandberg, Nick Bostrom, Sam Altman, Elise Bohan, and Randal Koene tend to cluster around their common retirement ages or late life stages, despite the calendar estimates ranging from the 2040s to nearer 2100. For some advocates, such as Ray Kurzweil, even their optimistic estimates are challenging – as a result they are working hard to live healthily and last long enough for digital immortality.

primates might be possible by the late 2040s.^[12] There is even further to go before non-invasive techniques could map neuron-level connectivity, let alone the full structure of the brain, but other researchers have identified plausibly adequate whole brain emulation by late this century.^[9]

There is a live debate about what level of physical brain mapping is necessary for mind uploading.^[15] Perhaps neuronal connections alone are insufficient, and we also need associated structures (glia, neurotransmitters etc.), subneuronal structures (microtubules, cell boundaries etc.), or system-level phenomena (electromagnetic fields, resonant structures etc.). However, technical barriers to such mapping are unlikely to prove insurmountable forever, in line with the incredible pace of recent progress in neuroscience simulation.^[10]

Ray Kurzweil, an optimist about mind uploading, describes a parallel in scanning the human genome.^[13] When the effort was first seriously discussed, sceptics argued genome scanning would take thousands of years by contemporary technology. Now, barely half a lifetime since those efforts, some 95% of a genome can typically be mapped within weeks on a commercial basis for individual consumers.

Part of the appeal of whole brain emulation is side-stepping the need to derive a full theory of consciousness. Provided you grant that consciousness emerges in some way from brain behaviour, whole brain emulation is not obliged to explain what about the brain actually delivers this result, nor does an ‘algorithm’ for consciousness need to be specified. Unfortunately, copying everything at the atomic level is out of reach even under the most ambitious of technology roadmaps.

Transhumanists have a practical solution to offer. We start with copying what we can, then interact with the resulting system to test if it actually seems to contain the personality and memory of the original person, including asking searching questions about its own conscious experiences. A strong result might be enough – at least for someone near death with no other options.

This paper raises a doubt against this practical ‘iterative map-copy-test’ solution. As I will explain, the possibility of adequate functional equivalence without *step-structure equivalence* means a part-way simulated brain might be wholly convincing in terms of the transhumanists’ tests but not yet implemented accurately enough to contain the desired consciousness.

Maintaining continuity of consciousness is a priority for most of the mind uploading protagonists discussed in this section. But what is it they want to maintain? I owe you a working definition of consciousness.

1.2. A working definition of consciousness

Would-be mind-uploaders do not necessarily share the same definition of consciousness. It is a famously contested concept.

In this paper, we will work with the common philosophical notion of phenomenal consciousness, as a ubiquitous aspect of human experience: the simple idea that there is an experiencing subject, some locus of awareness, or something that ‘it-is-like’ to be that system from the inside, as described by Thomas Nagel in 1974.^[22]

Without consciousness in this sense, there can never be a ‘you’ on the inside or any first-person perspective to speak of. In short, everything we have ever known is only known by us in virtue of our being conscious. With this expansive definition, it is obvious why mind-uploaders want to maintain it.

Without it, ‘they’ are no longer present in the new system in any meaningful sense. These descriptions of consciousness are somewhat circular, leading researchers like Pete Mandik to advocate abandoning them entirely.^[20] Nonetheless, they can help to isolate approximately the same concept in a shared dialogue, at least where we assume we have each experienced it first-hand.

Less circular definitions can come from contrastive methods, such as understanding consciousness as what goes away under general anaesthetic and returns when you wake up (or at least as one of the phenomena that go away). However, this definition is only approximate because we don’t technically know subjectively what happens under general anaesthesia. We simply infer we weren’t conscious due to our lack of memory about the anaesthetised state and due to the body’s inability to perform actions normally associated with consciousness.

A more rigorous definition builds on the example-based method of Eric Schwitzgebel.^[26] We identify consciousness as the lowest common denominator across all states we have ever directly experienced: the redness of seeing red, the feeling of joy, the sensation of cognition, imagining a concept, having a conversation, meditating into minimal states of awareness, and so on.

The lowest common denominator approach is introspectively challenging, as identifying such common features is hard to do rigorously.ⁱⁱⁱ It is also approximate because perfect extraction of a lowest common denominator would require experiencing every possible state. In practice, we must assume a certain breadth of experience is sufficient and that our interlocutors meet these standards. Nonetheless, with some effort, we can get to a sufficiently shared understanding to support constructive discussion.

A benefit of the ‘lowest common denominator’ definition is it helps to exclude certain features from explaining the minimal building blocks of consciousness, even if they might be usefully correlated with it or constitute sufficient causes under certain circumstances. If it is possible to be conscious in a state that lacks a certain feature (such as a minimum level of intelligence/creativity, self-awareness, a feeling of agency, a world model etc.), then that feature cannot be a necessary condition for consciousness.^[23] Rather, these features are phenomena that conscious experience is capable of incorporating and require a different sort of account within a theory of consciousness.^{iv}

1.3. Motivating physical functionalism

Now that we have a working definition of consciousness, we can return to the philosophical claims that mind-uploaders need to support their ambitions. First is ‘physical functionalism’. The ‘physical’ part is the claim that consciousness is identical with some relevant physical phenomena. Providing an

ⁱⁱⁱ Schwitzgebel provides a pessimistic take on the limits of ‘naïve’ introspection^[25]. However, it may be possible to improve introspection via training, e.g. Rademaker & Pearson where training on visual imagery improved metacognitive understanding of visual imagery shown in binocular rivalry tasks^[24], and Fox et al. where more experienced meditators had more accurate introspection over tactile sensitivity^[18].

^{iv} Consciousness science is not mature enough to rule out that the correct denominator would be phrased as a (partial) disjunction, rather than a single phenomenon shared across all instances. The necessary condition might in fact be an ‘any M of N’ rule, better operationalised as searching for sets of sufficient causes rather than a single necessary cause. Nonetheless, it remains reasonable to check simple rules first, being the approach commonly recommended in empirical research textbooks (e.g. Qualitative Comparative Analysis^[21]) and the current priority in neuroscience^[19].

exact definition of ‘physical’ can get tricky (see Hempel’s Dilemma as an example^[38]), but I’ll treat it as a working assumption for this paper.

To a first approximation, the ‘functionalism’ part claims that certain processes are conscious in virtue of the function they fulfil in a system, not the physical substances they consist of. Consciousness is a doing, not a being; a process, not a thing. However, there are many varieties of functionalism, discussed at length by Sydney Shoemaker^[37] in 1981 and in subsequent encyclopedia entries^[34].

For our purposes, the transhumanist test for consciousness can be seen as a particularly loose definition of functionalism, whereby any systems with identical input/output behaviour are functionally equivalent. What happens on the inside can be ignored so long as the results – the external behavioural responses – are the same. Consider how anything that ‘fulfils a social role of communication’ should count as communication; likewise for consciousness.

In this paper, we begin with a much tighter interpretation – call it strict functionalism – that is more faithful to arguments in favour of functionalism and provides greater safety for would-be mind-uploaders. Strict functionalism requires also the right internal causal mechanisms between the system’s inputs and outputs. The focus remains on what these mechanisms do, rather than what they are made out of, but we allow for the possibility that the detail of internal functions matter as well – at least until convinced otherwise.

By this strict definition, some functionalism seems to be an essential part of the picture, at least unless we invoke a well-tuned variant of panpsychism. If complex human-style consciousness has no function, it is hard to understand how natural selection could have recruited it, how we come to discuss it so much, and how it feels so important from the inside. It’s beyond our scope to recap the many debates about functionalism or to dwell on the subtle distinction between having a function and being a function, but it’s worth briefly pointing to some arguments against it.

Venerable thought experiments which critique functionalism include the following, allowing honourable mentions for arguments which can be adapted to target functionalism (e.g. critiquing the mainstream physicalist versions of functionalism given an original target of physicalism about consciousness):

- Mary’s Room, whether a scientist who knows all the physical facts about colour would still learn something new on seeing red for the first time;^[32]
- Philosophical Zombies, whether it is metaphysically coherent to imagine a person that is functionally identical to you but lacks subjective awareness;^[33]
- The China Brain, asking where consciousness would reside if the relevant neuronal function were emulated by large numbers of people signalling to each other via walkie-talkies (similar to the older Leibniz Mill argument).^[28]

There are also less well-known examples in recent years where physical functionalism is considered unlikely due to its seemingly absurd consequences, including the ‘Problem of the Many Minds’ where the human brain should be host to millions of identical causally redundant selves^[35] or Schwitzgebel’s argument that distributed systems like the US economy could be as conscious as we are^[36].

For the sake of argument, we will progress by assuming physical functionalism about consciousness. Without denying possible rebuttals, I will provide one influential thought experiment that defends it: the fading qualia argument.^[30]

Dave Chalmers asks us to imagine gradually replacing biological neurons with functionally identical artificial prostheses, one by one in a human brain. We can apply the same reasoning to any other cells in the brain or the body that might happen to matter for consciousness. Since the new cells are functionally identical, the human would continue to respond in exactly the same way throughout. The system's functional assessment of itself would also remain unchanged. However, if there is something about biology that matters for consciousness but is not captured in its function, at some point consciousness would fade or disappear. Such fading seems implausible if there are no other changes in internal/external system behaviour. Consider how conscious experience would disappear but the system either 'would not notice' or would act as though it did not notice, continuing to issue identical statements about the nature of its experience that now happen to be false.

Even if we grant the fading qualia argument, the question remains about which functions matter and how precisely they need to be defined. Reading the thought experiment closely, it applies strict functionalism: the internal causal mechanisms are exactly maintained. The new prosthetic cells are identical in every relevant aspect to the biological originals, without specifying exactly counts as relevant. As a minimum, they must capture everything about neurons that influences the spiking of downstream neurons, including the strength, speed, and nature of their onward connections, their electromagnetic fields for ephaptic effects, and their interaction with neuromodulatory chemicals. However, those who invoke the fading qualia argument typically assume it would translate naturally to computational interpretations of brain functions, including influential papers such as one led by Patrick Butlin and Robert Long (footnote 4 in their 2023 paper^[29]) and prominent researchers in AI, such as Geoffrey Hinton's LBC interview in 2025^[31].

Why do researchers feel so confident in computation that they can take an argument about functional equivalence and assume it supports computational equivalence, without even stating the leap, let alone defending it?

Some may feel that the rate of progress in modern AI needs no defending – every function or human capability will eventually be computable. Chess and route-planning fell long ago. Large language models are taking large bites out of creativity and reasoning. Consciousness will come soon enough. While persuasive in some settings, this argument is suggestive only, leaning on techno-optimism priors and uncertain trend-extrapolation. We will begin instead by making explicit the strongest argument that defends this leap to computational capability: the Church-Turing thesis that underpins Turing-equivalence.

Section 2. The nature of Turing-equivalence

2.1. The Church-Turing thesis

Turing-equivalence starts from Turing Machines.

An informal description of a Turing Machine is an unending tape divided into cells, where a finite set of non-blank cells each contain a single symbol from a finite alphabet. The tape is read one cell at a time by the tape head. The machine is always in one of a finite set of states (externally defined), with transitions to the next state determined by the current state and the tape symbol read by the tape head, compared against a discrete table of instructions for each combination. Example instructions might be to erase or write a symbol on the tape, to move the head, or to change state. The mathematics can be formally defined to tidy up some imprecision in Turing's original paper (e.g. the 7-tuple textbook definition^[41]), but the big picture remains the same.

Turing argued these machines can implement any ‘effective calculation’, i.e. any step-by-step mechanical problem solving in a discrete framework. In turn, any architecture that has equivalent functionality to a Turing Machine can also implement any such calculation provided they are allowed to become large enough.

Defining any Turing-equivalent architecture in the context of physical functionalism requires specifying both the system’s base algorithmic language and the physical mechanism that implements those steps. The two are closely related but not identical; the former corresponds to the smallest movements in the latter that have a computational interpretation in the target system. Together they form the key pivot by which a calculation becomes real and the results become available to a user. Example Turing-equivalent architectures include most programming languages implemented on digital computers (e.g. JavaScript, Microsoft Excel), various natural phenomena (e.g. DNA chemical reactions), and unusual examples that happen to have sufficiently well-defined state transitions (e.g. Conway’s Game of Life, Minecraft, humans communicating via semaphore).

Gualtiero Piccinini summarises the physical Church-Turing thesis in a single sentence: “All physically computable functions are Turing-computable”.^[44] It is important to note that the thesis is not a mathematically proved theorem, but rather a description about computation that is so widely accepted it is often treated as a definition.^[45] Its plausibility comes from the proven equivalence of various formal models of computation such as Lambda calculus and general recursive functions, the absence of counterexamples (despite efforts inspired by the acclaim that would await), and Turing’s original sketch about what step-by-step mechanical problem-solving entails in the human setting^[47].

For the purposes of this paper, the key insight is that all components of the Turing Machine are discrete (or can be treated as discrete with no loss of functionality). As a result, the binary, discrete logic-gate architecture of digital computers is no barrier to implementing any computable function. For consciousness to go beyond the Church-Turing thesis seems to require a function that cannot be computed – a mind-boggling concept. Mind-boggling it might be, but mathematical examples of non-computability are well-known and have been claimed to provide grounds for the non-computability of consciousness.

Non-computable phenomena are effectively those which cannot be resolved directly by step-by-step mechanical problem solving. However, mathematical reasoning can still infer insights about such phenomena. Firstly, we can mathematically construct and manipulate objects that do not ‘fit’ on a Turing Machine in any finite time. For instance, computable but irrational numbers can never be written out in full on a Turing Machine, including numbers that appear central in mathematics, physics, and nature – and might therefore be relevant to consciousness (e, ϕ , π etc.).^v We can also construct explicitly non-computable numbers, such as Chaitin Omegas and most real numbers. Secondly, many proofs exist that draw on non-computable mathematics (e.g. the Tiling, Halting, and Totality problems; Gödel’s incompleteness theorems). Thirdly, there are ideas based around hypercomputation: theoretical systems that could conduct computations beyond Turing Machine limits, although the physical feasibility of such systems is contested.

However, these same limits can be argued to apply to human cognition as well. We also cannot write out π in full in our minds, even though we can manipulate symbolic representations of the concept of π in mathematical reasoning (as can Turing Machines). Humans also cannot solve Tiling Problems in

^v However, these numbers can be emulated to arbitrary precision on a digital system, including beyond the Planck length in a given physical implementation. Any functional relevance of further physical precision in an analogue system would seem to go beyond our current understanding of physical laws.

the general case, only prove facts about them and construct solutions in individual cases (as can Turing Machines). The claim for human supremacy rests instead on the meta-computational insights we draw from these proofs, with famous arguments provided by Roger Penrose, the Nobel Prize-winning physicist.^[43] For instance, having followed Gödel's diagonalisation argument, we can step outside the specific mathematical formalisation we used to follow it and 'see the truth of it' as applied to any and all formalisations (technically the Turing Machine only proves this truth within one or a succession of specific formalisations). Such arguments are possible in principle but have held little weight with mainstream neuroscientists and computer scientists to date.

In this paper, I explore instead the limits of Turing-equivalence via the concept of 'step-structure', remaining agnostic on the issue of non-computability.^{vi}

2.2. Turing-equivalence is powerful but does not extend to 'step-structure'

From a mind uploading perspective, the equivalence of interest is the brain as a physical system (whose physical operations are assumed sufficient in some way for human-style conscious experience) and a programmable digital computer (e.g. Python code combined with an adequate number of physical NAND gates and a clock function). But what does it mean for a function to be implementable in a Turing-equivalent way across both architectures?

Formally by Turing's original definition, all that is required is input/output equivalence, including halting (or failing to halt) in all the same calculations. Turing's original argument centred on emulation, not identity: a human clerk could emulate any mechanical method – step-by-step – with pen and paper, so a machine that can do this is universal with respect to mechanical calculating methods as humans understand them. This meets the loose definition of functionalism applied in the transhumanist tests for mind uploading, but not the strict definition implied by the fading qualia argument.

Turing-equivalence does not formally require the same internal causal logic, i.e. which early steps in the calculation feed into which later steps. For instance, provided the multiplication always produces the same result, it does not matter if the multiplication is achieved by a repeated addition algorithm, long multiplication (e.g. a look-up table memorising simple integer multiplications up to 10x10, decimal place value representation, and partial product addition), Karatsuba's algorithm, or Fast Fourier Transform multiplication.

Various theories of consciousness do, however, make explicit requirements about causal logic. For instance, Recurrent Processing Theory requires feedback loops.^[49] If this theory is true, an algorithm that is input/output equivalent but lacks loops would not be conscious under the theory, despite being behaviourally identical from an observer's perspective. Adrien Doerig and others explained how feedforward representations (i.e. without feedback loops) can always be constructed or 'unfolded'

^{vi} Many other arguments arguably question the possibility of digital computer consciousness based on the supposedly absurd implications that seem to result. Examples include:

- the Slicing Problem, which identifies arbitrary consciousness-multiplying exploits^[40];
- the idea of Fractional Qualia which permits hard-to-imagine experiences like hearing 0.5 of a sound^[39];
- the Olympia thought experiment implying that consciousness could depend on whether an inert part of the mechanism happens to be present (even if it is never used)^[42]; and
- triviality arguments that diverse conscious experiences should be present in many physical systems^[46].

Any of these arguments being valid would also support this paper's concerns about digital architectures for mind uploading.

from any artificial neural network^[48], so this possibility is particularly pertinent for contemporary theories about the human brain. Particular causal patterns are also needed for such prominent theories of consciousness as Global Workspace Theory and Attention Schema Theory.^[49]

The requirement for specific causal logic might already seem enough to defeat Turing Machine equivalence as a *guarantee* of preserving consciousness under functionalism across different architectures. (Of course, algorithms in different architectures might happen to have the necessary causal logic, but it cannot be safely assumed without checking). Not so fast. Turing Machines are not so easily limited. These machines can also emulate any arbitrary internal causal logic, including feedback loops – simply because we can take any individual logical statement within the calculation, interpret it as a local input/output mapping, and then emulate it. If a feedback loop is needed, I simply emulate that feedback loop on a computer. An input/output-identical ‘unfolded version’ without feedback loops still exists in theory (and can be built), but I don’t have to use it. Instead, I use the version whose causal structure contains a logical feedback loop.

Turing-equivalence can therefore mimic the internal *causal logic* of any computation, even if the definition does not formally require it. This is powerful, but it does not go all the way towards the identical *causal mechanisms* required under strict functionalism. Specifically, Turing-equivalence does not guarantee the specific nature of any individual computational step in its architecture.

Consider again the wide range of Turing-equivalent architectures and the different ‘computing units’ they build from. For instance, we might have a biological neuron which takes in thousands of inputs to drive each individual decision about whether to activate or not, compared to a binary logic gate that takes in only two inputs. As another example, perhaps the individual inputs to the base computing unit can take only two values (0 or 1), three values (ternary), or even any computable number between 0 and 1.

From a causal mechanism point of view, there’s a significant functional difference between analysing just two inputs at a time compared to analysing thousands. The difference is in mechanism rather than logic. Enough binary logic gates can perfectly emulate the causal logic of a biological neuron, in that it is possible to identify an abstract mapping that is one-to-one with the neuron’s causal logic, but the step-structures of the causal mechanism and its base algorithmic language remain binary-input only.

When written out in full detail in the base algorithmic language of a given Turing-equivalent architecture, these causal mechanism differences translate into differences in algorithmic specification. Some computations are trivial to specify in one Turing-equivalent architecture, but require reams of code in another. Just ask any battle-scarred programmer about floating point errors and cases where $0.1 + 0.2 \neq 0.3$. In a strict mechanical sense, these are different functions that simply happen to map to each other’s logic – and often require rounding for an exact mapping. However, Turing-equivalence draws its force from this lack of mimicry. The whole point is that changing computational steps between different mathematical formalisms or physical implementations need not change the input/output behaviour of a computation or its individual logical operations.

The required principle for this paper can be summarised as the Step-Structure Principle: Turing-equivalence does not extend to step-structure, where the step-structure of a given calculation refers to the number and nature of inputs/outputs in each computational step in an architecture’s base algorithmic language, as well as the aggregate metrics that flow from these parameters, such as the total number of steps in a calculation or the step with the largest number of inputs. The statement is mathematically uncontroversial (Turing would not see it as a critique of his thesis); the impact lies in drawing attention to it explicitly and motivating why it might matter.

Note also that this definition creates a link to the physical implementation of a computation, but is not in itself physical. We remain focused on the computation taking place, not the material out of which the computing object is made. This is appropriate, given our assumption of physical functionalism. The target theory of consciousness is substrate-neutral, not “unphysical”. Whatever substrate is chosen, the algorithm must ultimately be physically implemented in the Turing-equivalent architecture for consciousness to result, and this physical implementation introduces new considerations that draw on step-structure.

Section 3. Why ‘step-structure’ might matter for human-style consciousness

The Step-Structure Principle states that step-structure is not necessarily carried across Turing-equivalent architectures – but so what? Why might this matter for human-style consciousness? Our scientific and philosophical uncertainty about consciousness limits any definitive claims, but I will sketch three reasons why step-structure might matter, grounded respectively in the computational cost in complex systems, phase transitions aligning with evidence in empirical neuroscience, and the phenomenal binding problem. These three reasons are not interdependent; any one being successful would be sufficient for the critique of mind uploading trajectories. The reasons are uncertain to different degrees, but collectively they create sufficient space to take the critique seriously and invest in evaluative research.

3.1. Computational cost in complex systems

Turing-equivalence may not care about computational efficiency, but evolution does. Much has been made about the human brain’s abilities at a mere 20 watts on average, compared to the energy-hungry nature of today’s AI models.^[53]

A high-level algorithm with some relevance for evolutionary fitness might be implemented in multiple ways; faster or lower energy cost implementations would naturally be favoured where both are similarly accessible in the evolutionary environment. To illustrate the point, the algorithm might involve deciding whether to flee a possible predator and how much energy to invest in doing so. Slow decisions mean death. Perhaps the algorithm can be implemented firstly via fast, efficient feedback loops or secondly via a fully unfolded feedforward neural network with identical input/output behaviours. If recurrency is a necessary (even if not sufficient) condition for consciousness, nature might naturally adopt the first implementation for its effectiveness, happening to lay down a path towards complex human-style consciousness at the same time. Meanwhile, humanity might choose to build the second for unrelated reasons.

You might ask about a possible third implementation in a digital computer: an even more sprawling network that mimics the feedback loop logic of the first. Even if it needs millions of logic-gate computational steps for each loop, the emulation remains in place: feedback loops exist and the exact same conscious experience would manifest as in the evolved scenario (so functionalists claim), depending on any other necessary conditions. It might not be evolutionarily viable, but it’s still technically viable. Or is it? It would depend on the algorithm.

For sufficiently complex algorithms operating on a digital computer architecture, it is conceivable that exact mimicry would require an implausibly large calculation, translating into such physical scale and such levels of error correction that the resulting computer might simply be physically or technically

impossible. Being philosophically conceivable is little consolation to today's transhumanist who is confined, despite their optimism, to our physical universe.

Would such a situation ever be necessary for consciousness? In the absence of a detailed theory, we cannot be certain either way, but consideration of complex systems opens the door to its possibility.

It is well-known that present-day computers cannot simulate physical systems beyond a given level of complexity and accuracy, well before invoking any theoretically unattainable limits such as the exact value of π or non-computable numbers. Examples include many-body interaction problems, chaotic systems that are sensitive to their starting conditions, and large quantum entangled systems. Even looking towards future computers, it is not hard to identify thresholds where simulating the system is only feasible by actually implementing the system, in line with notions of computational irreducibility explored by Stephen Wolfram.^[52]

Depending on our assumptions about future computational capabilities, quantum systems with computational irreducibility likely occur at surprisingly small scales, e.g. exact simulation of several hundred strongly entangled particles becomes hard to achieve even on universe-scale computers, as the quantum information scientist Seth Lloyd argued in the early 2000s.^[50,51] Thermodynamic issues around stochastic noise and error correction might narrow the limits even further in practice.

It might be that human-style consciousness requires so complex an algorithm operating with such precision that the only realistic computers capable of implementing it require biological-style components. Such components effectively serve as analogue computers refined for specific operations, unlike the more expensive but flexible nature of programmable general-purpose digital computers.

3.2. Phase transition motivations

The view we are scrutinising holds that most simple types of computation are not conscious. It is complex computations that are conscious, whether the relevant complexity lies in specific functions, brute scale, or both. If we take consciousness seriously as a phenomenon – as do the mind uploading protagonists hoping to preserve it – this is a serious form of emergence. An entirely new phenomenon emerges: awareness from non-awareness, first-person perspective from third-person interactions, subjectivity from objectivity. We can debate what is required for such emergence, but a plausible place to look is phase transitions in nature, where dramatically new behaviours can emerge on the other side of a phase transition.

Phase transitions transmute states before and after combustion; between ice, water, and vapour; and between planetary masses, stars, and black holes. Phase transitions in nature have a key thing in common: scale matters, in the sense of 'spatiotemporal density' (or 'intensity') of the system. Are the relevant structures or activities sufficiently 'compressed' together in space-time or are they too widely dispersed for the relevant phase transition to occur? Structural organisation and other factors might also matter, but this spatiotemporal density still plays a necessary role.

For instance, sustained combustion from oxidation relies on the frequency of reaction events taking place beyond a context-dependent threshold, which can be interpreted as the 'temporal density' of those reactions (i.e. how many reactions per second) within a given volume of space. Ice, water, and vapour transitions result from the interplay between molecular kinetic energy (temperature) and the spatial concentration of molecules (density), which determines the strength of intermolecular

interactions. Stellar fusion and black hole formation require simultaneous spatial concentration of mass beyond particular thresholds, providing necessary conditions for their phase transitions.

Reliance on spatiotemporal density differs from computation, where the outputs of a prior computing step can be as distant or as delayed as you might wish, so long as the outputs are accurately preserved and entered into the next step. An AND gate that computes over light years from one galaxy to the next is still a Turing-exact emulation of a near-instantaneous AND gate conducted on your computer. By contrast, emulating density does not generate physical density: actual spatiotemporal coincidence matters.

If human-style consciousness requires a phase transition and spatiotemporal density matters for phase transitions in a physical universe, then step-structure is plausibly relevant. Perhaps our consciousness requires a minimum number of computational steps within a certain physical space. Perhaps it requires a spatiotemporal concentration of information processing (e.g. number of bit flip equivalents in a cubic centimetre) or a minimum number of inputs/outputs for the physical mechanism implementing a calculation. These are hypotheses to be tested, not claims to be decided from an armchair.

Just as a certain minimum amount of mass needs to be compressed into a small volume of space for stellar ignition, perhaps consciousness ‘ignition’ also requires a certain minimum amount of information processing to be compressed into a small volume of space.

Why should we believe that density or intensity is part of the picture for human-style consciousness, even if only as a necessary condition?

Our best evidence from empirical neuroscience is consistent with such beliefs and several theories invoke phase transition-like concepts explicitly.

A slew of neuroscientific evidence identifies that stimuli below a certain threshold trigger neuronal activity but do not enter into conscious awareness, whereas above-threshold activity is associated with such neuroscientific markers as increases in synchrony, broadband gamma amplification, and a late P3 wave.^[55,56] There appears to be a sharp qualitative change in experience and non-linear changes in brain behaviour associated with transiting a threshold, just as we observe with physical phase transitions. The evidence on complexity thresholds is considered strong enough that it has been used for gauging consciousness in brain-injured patients and triaging appropriate treatments. For instance, clinical judgements are supported by the Perturbational Complexity Index, which extracts a complexity score from EEG response data following brain stimulation.^[54,57]

Inspired in part by this type of evidence, Global Neuronal Workspace Theory explicitly describes a non-linear “threshold for global ignition” that leads to consciousness, even if the exact nature of that threshold is not yet fully understood.^[49,55] As a second example, the Theory of Neuronal Group Selection describes thresholds for consciousness based on, for instance, “extensive reentrant interactions” with appropriate levels of causal density and relevant complexity.^[58] As a third example, the Dynamic Core Hypothesis predicts that “complexity of the dynamic core should correlate with the conscious state of the subject”, such that a non-conscious human brain would have below-threshold complexity.^[59] Complexity thresholds in a spatiotemporally constrained space (human brain processes) is exactly the kind of density phase transition in scope.

Even where these theories are interpretable as identifying correlates of some underlying behaviour that actually drives consciousness, the evidence behind them is equally consistent with the claim that

this complexity/density threshold is a necessary condition alongside those underlying causes. Further research would be needed to interrogate this key distinction.

3.3. Phenomenal binding motivations

An entirely different type of motivation is philosophical and phenomenal in nature. Our conscious experience is capable of containing multiple bits of information at the same time. Introspection on the visual field or on working memory trivially identifies minimum levels of complexity in a single experience. Even if some of that complexity is an illusion, if consciousness is not perfectly unified, or if it only appears unified when we turn attention to it, there remain moments of experience that present multiple bits of information to us at the same time, sufficient to demand an account for the phenomenon.

The ease with which complex information can be processed by machines can make the phenomenal binding problem seem easy to solve at first glance (consider an Excel spreadsheet or the feats of large language models). However, complex information processing and complex experience have fundamentally different requirements. Even if the same mechanism might underpin both, explaining how that mechanism delivers both requirements remains a puzzle within philosophy and neuroscience.^[65,66,67] Information processing often proceeds by breaking down information into its constituent parts and operating on each piece separately over multiple different steps. Phenomenal binding requires ‘phenomenal simultaneity’ across that complex information, maintaining it all at once rather than reducing it.

One factor behind the challenge of solving ‘phenomenal binding’ is that our models of physical behaviour and computation tend to separate phenomena into constituent steps connected through time, rather than connecting a targeted subset of them in the same experiential moment. It is possible that the relevant binding mechanism does operate across an extended period of time, i.e. units of information are perceived as simultaneous even though they are not physically simultaneous. However, it proves surprisingly hard to find a plausible cross-time mechanism that does not ‘bind too much’, i.e. would not conflict with evidence on information processing in the brain that is outside our conscious awareness but is nonetheless connected across time to information that is consciously experienced.

There is no established solution to phenomenal binding, with proposed mechanisms remaining controversial and many researchers focusing instead on easier topics. Proposed mechanisms typically favour actual or approximate physical simultaneity as a solution, with candidates including shared resonances in an oscillating structure^[60], certain structural features of brain-based electromagnetic fields^[61,63], the collapse of quantum entangled systems^[67], and the novel definitions of existence and causality that are twinned with emergence metrics in Integrated Information Theory^[64,65]. It remains an open question whether sophisticated phenomenal binding can emerge through computational complexity alone, although possibilities are being explored such as the structure of Bayesian reasoning and coarse-grained informational closure.^[62]

The key link to this paper is the possibility of leveraging computational step-structure to create a natural notion of simultaneous complexity. A single physical act that implements a given computation feasibly interconnects all the inputs’ information to that computation, by virtue of holistically integrating them to generate the output. Where such physical acts provide inputs to conscious experience, we might expect the computation’s inherent unity to manifest in a single experience, especially where the single physical act is in fact a unitary object in a physically meaningful sense.

Physical computing mechanisms can vary widely in their step-complexity for a single computation. A particularly complex mechanism might provide the missing explanation for phenomenal binding and unlock the sophisticated experiences of interest for mind uploading. Digital computers under computational functionalism would, however, let us down. Such computers explicitly implement Turing-equivalence via physical mechanisms that are designed to have very limited complexity: think idealised logic gates with two binary inputs and one binary output. A single set of two binary inputs cannot represent the individual moments of complex experience that we hold so dear.

Contrast binary logic gates with the single physical step computation possible in an optical computer or a quantum computer. In optical computers, a single operation might take place on a photon's phase to encode a continuous variable – vastly more information than a binary bit. Electromagnetic field mechanisms can be exploited in various ways to perform complex calculations in minimal physical operations, such as a single lens implementing Fourier transforms or diffractive deep neural networks performing matrix multiplication. Quantum computers can also embed very large amounts of information into single transformations of a unitary quantum-entangled structure.

Viewed through this lens, human-level phenomenal complexity is within reach for even small-scale systems. In both cases, current technology and physical laws place severe limits on the information users can practically extract about the computation's end result (e.g. Holevo's Bound), as well as limits on encoding precision. However, these limitations relate to exploiting the physical mechanism for external users' calculations, rather than preventing the mechanism from unfolding on its own terms: the rich information of its inputs is still processed simultaneously in a single physical operation.

3.4. An aside on temporal continuity

You might reasonably ask about temporal continuity under this view of phenomenal binding.^[70] Physically complex systems might compress millions of bits of information into a single physical implementation of a single computational step. All that information may suffice for the phenomenal complexity we experience in a given moment, but any subsequent computational step would be a separate phenomenon. The second step might have its own internal unitary nature, but it would lack any binding mechanism that joins directly with the contents of that previous step, because the claimed binding mechanism requires inclusion within a single shared – and perhaps physically irreducible – step.

None of this prevents us from feeling the passage of time. The physical computation likely takes some time to unfold, but even without this, it is trivial for information about the past to be time-stamped and interpreted accordingly in a simultaneous analysis. Likewise, memories about a past narrative self can easily be incorporated into isolated present moments. However, this view of phenomenal binding does mean the common impression of temporal continuity is an illusion. Plausibly, the conscious self who finishes reading this sentence is different to the one who began it – and likely different to the one who went to bed last night.

This view of self-continuity may seem shockingly out-of-sync with the commonsense self-models with which we navigate day-to-day existence. It need not, however, be philosophically or spiritually shocking. In fact, it follows a long tradition of insight about the illusory self under Buddhism^[68], the personal identity paradoxes explored by Derek Parfit^[73], and introspection on Ship of Theseus arguments. Would-be mind-uploaders have little choice but to embrace a view of this type, given the

well-known puzzles about what happens to your ‘single’ conscious self if an uploaded mind is duplicated.^[74]

What would we owe such disconnected future selves? Well, perhaps this is an irony of ethics: the Rawlsian Veil is not just a reasonable guide to ethical behaviour towards others, it is in fact the underlying physical truth of our conscious minds and our duties towards our successor selves.^{vii} Options also exist for countering existential angst that might be felt about such a view, including open individualism^[72], cosmopsychism^[69], and valence realism^[71], but these are ideas for another time.

Section 4: Pathways to empirical testing

All I have done so far is raise the possibility that step-structure matters. Establishing some confidence over whether it actually does matter will need the help of science.

All tests for consciousness ultimately trace back to self-report – and face the inevitable limitation that self-reports might be inaccurate or misleading. Even so-called ‘no-report’ paradigms, such as looking at brain behaviour, are ultimately anchored in the fact that such brain behaviour has correlated with self-report in previous studies. However, while absolute proof regarding consciousness might be out of reach, it is still possible to make progress within these constraints. Consider the recent adversarial collaborations in lab experiments testing theories of consciousness head-to-head (such as the Cogitate Consortium) and the accumulation of empirical evidence that generates inspiration, requirements, and constraints on candidate theories.^[76,79] In this spirit, let me sketch two high-level pathways to testing the central ideas in this paper.

The systems we want to differentiate are input/output identical for behaviours (but identical in all respects), so we’ll need to be careful with the tests we use. And accept tests that build confidence one way or another, rather than provide an exact result. Addressing at least part of the risk is worthwhile, even if philosophical certainty on all fronts remains out of reach.

The first pathway turbocharges the current research programme for neural/physical correlates for consciousness, anchoring it in identifying phenomenal binding mechanisms that draw boundaries around complex conscious experiences and differentiate unconscious from conscious information processing in the brain. Future brain imaging techniques can be used to investigate theorised binding mechanisms – ranging from synchrony to electromagnetic field topology, from neuronal ensemble activation loops to coordinated quantum entanglement – all of which make different predictions about which physical structures map to which complex experiences. With a database of physical structures in the brain mapped against reported experiences in different moments, we can see which line up and which do not.

We should expect mapping techniques to narrow the field of plausible theories, but they may not identify a unique candidate (or set of equivalent candidates for mind emulation purposes). If so, we will need to move into perturbation techniques. As we intervene in the brain to hold certain structures fixed while varying others, greater care is needed because of the possibility that functional equivalents or near-equivalents might lack human-style consciousness. A pragmatic approach focuses on only the tiniest of experimental adjustments from the natural domain in which we’re willing to trust it, i.e. the evolved environment of today’s human brain. The larger the adjustment the less we can trust it. Example experiments include voltage clamps to control neuronal firing outside of their input

^{vii} The philosopher, John Rawls, described making moral decisions under a veil of ignorance where you don’t know what role you would have in the situation being decided on.

structure, differentiating mechanistic and counterfactual views of computation, as well as opening up routes to explore certain electromagnetic field theories of consciousness.

Another example experiment is separating brain modules (and perhaps individual neurons) from each other, artificially transmitting their interconnecting signals across different distances and investigating any changes in brain behaviour, phase transitions, and self-report. An information density perspective expects no difference via artificial transmission under negligible differences (e.g. fractions of a millimetre) unlike under larger distances. If there are such differences, they will also help us specify new quantitative criteria for mind uploading machines. Although highly challenging, these experiments are technically feasible if neuroscience continues to progress. They could, for instance, build on the scattered brain experiment protocols outlined by Albert Gidon and his fellow neuroscientists in the early 2020s.^[77,78]

The second pathway is empirically-informed theory development. For instance, we can theorise about which physical correlates have the right features to underpin phenomenal binding. We can analyse the computational cost of simulating different levels of precision. Theory might already help rule out some positions and inform a sensible level of confidence, given what we know so far.

We can explore the possibility that physical density of information processing drives a functional phase transition by using deep mapping and theorising about the brain to specify that phase transition. What is the underlying phenomenon becoming more dense, at what threshold does something change, what is that phase transition, and why might it be plausibly relevant for consciousness and broader brain function. For instance, even if information processing can remain the same in principle, it might be that the post-transition state enables faster or more energy efficient processing – a huge evolutionary advantage, even if it looks the same in Turing Machine equivalence. If we can't find such phenomena, the possibility starts to look less likely. Experimental insights would help theorists identify specific phase transitions rather than gesturing to their likelihood given indirect evidence from neuroscience. Specifically, the order parameter, critical points, and underlying physical phenomena all need to be described and motivated before a phase transition theory can be convincing.

All of this research is exceptionally challenging, but this should not dent our ambition. Neuroscience has already made incredible progress and AI-enabled acceleration might take us further, faster.

Large investments have unlocked incredible scientific progress in the past, such as developing quantum physics via particle accelerators, sequencing the human genome, landing on the moon, or improving cancer survival rates. The availability of funding is, as ever, more a matter of social priorities than actual resources. To single out two examples somewhat unfairly: companies spent over US\$200 billion on TV/film content in 2024^[75], and clubs spent over US\$10 billion on professional football salaries in Europe alone in 2023^[80].

The benefits of progress are compelling. If safe mind uploading is not motivation enough, we can also expect to find new edge pieces that help solve the eternal jigsaw puzzle of consciousness.

Conclusion

This paper has challenged an argument that seems to guarantee the philosophical viability of human-style digital minds, whether new minds birthed in software or individual human minds uploaded to a machine. The result weakens the guarantee into a conditional possibility – something to be checked rather than assumed.

The original argument I am challenging can be summarised via two premises. First, physical functionalism: consciousness is a function of what the brain does as a physical object. Second, Turing-equivalence: any function can be adequately implemented on any Turing-equivalent architecture, including digital machines. As a result, any large enough digital computer with the right software is capable of instantiating any conscious entity/experience. That software specification might be unknown today, but we know that any possible algorithm is within reach. With perfect enough copying, we could even upload minds without knowing the algorithm.

Allowing the first premise for the sake of argument, I add a caveat to the second, given the Step-Structure Principle that limits Turing-equivalence. The consciousness function can only be safely implemented on Turing-equivalent architectures, *if the step-structure of that function does not matter*. As a result, we cannot guarantee that digital computers suffice until we have checked whether step-structure is relevant for human consciousness. What happens next for mind uploading depends on what we learn. Take three possibilities identified so far via the reasons step-structure might turn out to matter.

Via the first reason, if the relevant physical mechanisms that compute consciousness algorithms are computationally irreducible (or impractical to implement otherwise), then artificial consciousness will be possible only where it explicitly draws on those physical mechanisms. Nonetheless, the class of adequate physical mechanisms may not be solely biological. It depends on what algorithms are involved. Mind uploading might be possible into an analogue or hybrid computer. Existing computers in early-stage R&D are already leveraging analogue components, such as neuromorphic, biological, field, and quantum mechanisms. These might suffice or they might not, we simply do not know enough at this stage.

The second suggested reason for step-structure is that the ignition of consciousness might require a phase transition based on some intense spatiotemporally-localised level of information processing, whether as a computational act or via its physical implementation. While the computer's formal information processing logic would remain unchanged, other functional aspects of the system might differ, such as those based on energy or thermodynamic behaviour. Digital computers might do the trick, depending on the details. It would likely require a carefully-designed chip structure running on a physically contained system. We're less likely to get there with algorithms distributed in cloud computing or efficient calculations parsed into parallelised micro-steps.

Finally, if a phenomenal binding mechanism is required, then complex conscious experiences within a machine would also require complex physical mechanisms, such as machines incorporating analogue components. Precise phenomenal binding mechanisms are yet to be found, but existing theories already provide some signatures to look for in the human brain and future theories should be encouraged.

This paper's quarrel is less with the transhumanist dream and more with their technology trajectory.

The transhumanist trajectory assumes that good enough brain mapping and big enough digital computers are all we need. This assumption is not secure. More caution is needed, especially as Turing-equivalence *can* guarantee input/output-equivalence. Even if step-structure matters for human consciousness, we would not necessarily see the difference in the external behaviour of an uploaded mind (or even in its internal causal logic). The 'new you' would have your memories and interact with others just like you. By the usual test for mind uploading, it has been a great success – likely a very

profitable success for someone. But there would be nothing on the inside – no subjective experience, no-one to feel what it's like to be you.

This is exactly the greatest risk of the current trajectory. Our test for success smuggles in an insecure assumption. It looks like it works when it might not. One-by-one we might follow our early-adopter friends into what looks like a natural extension of goals we already pursue: better health, longer friendships, new challenges, and self-improvement. One-by-one we switch off our inner light until our civilisation has become an empty theatre; our once-vivid stories acted out by robots on a dark stage.

Mind uploading might seem like science fiction, but serious people are investing in it today. It might seem unnatural, but it would simply be the latest in a long line of technologies humans use to pursue common goals. Consider how the technical pathway might proceed step-by-step, reducing any understandable anxiety about the whole transition. Just as mechanical aids have extended from external devices (such as glasses) to internal devices (pacemakers and cochlear implants), so might cognitive aids extend from external tools (books and online calendars) to brain implants that enhance memory or cognition. As brain implants blur the boundary between cognition in wetware vs software, remote robotic implants could blur the boundary between actions taken with our physical bodies vs our distributed systems. Over time, it may be easiest or cheapest to work solely with the latter.

The first full uploads that abandon their original meat bodies might only be the elderly or those near death – and who would oppose them? But as the rest of us see them thrive, a medical last-resort might fast become a trendy first-choice.

Unfortunately, early uploads are where the risks are sharpest. Early uploads will most likely take place before full brain mapping has been completed. After all, if you're about to die, why not take a risk with a simulation based on neuron-to-neuron connections alone? You don't have the luxury of waiting for more detailed chemical or ephaptic mapping of your brain. With the likelihood of decades in delay for more detailed mapping and order of magnitude increases in cost, humanity will choose the earliest solution that seems to work, even though that is the solution least likely to be robust to the step-structure assumption. The corporate first-mover advantage might be the beginning of the end for consciousness.

If the technology can deliver it, the transition to wholly uploaded minds is a plausible pathway for humanity – and it might take us to magical places. But there is a risk it happens too fast, before we've got the foundational research right. Those who would live forever must do their homework first. Mapping the brain is not enough, we need to find and test the source of consciousness before anyone should be confident in mind uploading.

Perfect certainty may be out of reach (as for many philosophical questions), but our evidence base for decision-making can be vastly better. Good research can explore the step-structure assumption.

Perhaps we will learn that digital minds can never truly live, while analogue minds truly can live forever.

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